Quantum 0-infinity law: divergence of the series of ratios is equivalent to totality of the random orthonormal sequence (Theorem 2.1)

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/sigma-2022-012-a-quantum-0-infty-law/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span is the author Theorem environment theo:maindense (source lines 129-135); its proof is split by the author into two named proof environments (source lines 311-319 and 373-395, opening words 'Proof the first assertion' and 'Proof the second assertion' retained verbatim) and both are delivered as proof spans in the author's order.

Editorial note. The definition and probability-setup contexts (source lines 94-120 and 127) and the U/V/W setup (181-188) sit before the statement; the sibling Lemma 4.2 (243-252) and Corollary 4.5 (289-295) are theorem-like contexts placed after the statement so the card keeps 2.1 (both print locally renumbered). Context line 98 is excluded because it cross-references the conclusions section, outside the card; the measure context stops at line 107 for the same reason.

Editorial note. No bibliography fragment: no cite commands inside the delivered spans or contexts.

Editorial note. Printed number anchored to the published PDF (SIGMA 18 (2022) 012, read from the stream-extracted text of sigma22-012.pdf, census pdf_anchor_evidence): the theorem family prints 2.1, 3.1 and 4.1-4.8 with the Corollary at 4.5 sitting mid-sequence between Lemma 4.4 and Lemma 4.6, confirming the sectionwise shared master counter Theorem declared in the author's newtheorem block (source lines 34-41). The pinned tex has five sections plus appendix A and its structure matches the published PDF.

Editorial note. Theorem-family numbers inside the card are locally renumbered; every cross-reference is a symbolic ref and resolves inside the card. The source displays are unnumbered, so no equation numbering surface exists.

Editorial scope. Single environment card cut from the SIGMA article doi:10.3842/SIGMA.2022.012 (sigma-2022-012-a-quantum-0-infty-law); statement and proof are the author's own text at source lines 129-135 and 311-319 and 373-395.

Allowed standard prerequisites (editorial).

- Matryoshka and probability setup. Delivered in context (source lines 94-120): the increasing dimension sequences, the n-Matryoshka F, the m-Matryoshka E in F with E_k in F_k, the Grassmannian measure mu and the recursive probability measure P on the set M of Matryoshkas, with the filtered space (M, algE, (algE_k), P).. Load-bearing: fixes every object the statement and both proofs quantify over.

- U/V/W decomposition setup. Delivered in context (source lines 181-188): the orthogonal projections U_k(v), W_k(v) and V_k(v) and the three-term decomposition of v and of the Hilbert space.. Load-bearing: W_k(v) and its projections are the quantities whose large-k behavior both proofs track.

- Conditional-expectation recursion (Corollary 4.5). Delivered in context (source lines 289-295): E(norm W_k(v)^2) = E(norm W_{i-1}(v)^2) times the product of (1 - r_j), bounded by norm(v)^2 times the same product.. Load-bearing: proof 1 reads the divergence case off it (source line 318) and proof 2 builds the lower-bound vectors on it (source line 376).

- A.s.-to-zero characterization (Lemma 4.2). Delivered in context (source lines 243-252): W_k(v) converges almost surely to zero iff the expected squared norms converge to zero.. Load-bearing: converts the expectation statement into the almost-sure totality claim (source line 316).

Let $\underline{n} := (n_k)_{k\in\mathbb{N}}$ be an increasing sequence $0 = n_0 \leq n_1 \leq n_2 \leq \cdots$ of integers with $\lim_{k\to\infty} n_k = \infty$. An $\underline{n}$-Matryoshka $\underline{F} := (F_k)_{k\in\mathbb{N}}$ is an increasing sequence $\{0\} = F_0 \subset F_1 \subset F_2 \subset \cdots$ of subspaces of $\mathcal{H}$ with $\dim F_k := n_k$ for $k \in \mathbb{N}$. An $\underline{n}$-Matryoshka $\underline{F}$ is called total (in $\mathcal{H}$) if $\cup_{k\in\mathbb{N}} F_k$ is dense in $\mathcal{H}$.

In the framework of numerical computation of low-lying eigenstates, F_k would be the kth trial space, the space in which the first k levels are to be found.

We also fix an increasing sequence $\underline{m} := (m_k)_{k\in\mathbb{N}}$ of integers such that $m_k \leq n_k$ for $k \in \mathbb{N}$ but $\lim_{k\to\infty} m_k = \infty$. We set $(\delta m)_k := m_{k+1} - m_k$. An $\underline{m}$-Matryoshka in $\underline{F}$ is an $\underline{m}$-Matryoshka $\underline{E} := (E_k)_{k\in\mathbb{N}}$ such that $E_k \subset F_k$ for $k \in \mathbb{N}$. In the sequel, if $\underline{E}$ is an $\underline{m}$-Matryoshka in $\underline{F}$, $E_k^{\perp_F}$ denotes the orthogonal complement of E_k in F_k (not in $\mathcal{H}$).

We denote by $M_{\underline{F},\underline{m}}$, or simply M when there is no risk of confusion, the set of $\underline{m}$-Matryoshkas in $\underline{F}$. We recall that if F is a finite-dimensional Hilbert space[1] of dimension n and $m \leq n$ there is an unique unitary invariant probability measure μ_F^m on the set of m-dimensional subspaces of F (a Grassmannian which we denote by $\mathbb{G}(F, m)$).

We use the measures μ_F^m to endow the set M of $\underline{m}$-Matryoshkas in $\underline{F}$ with a probability measure $P_{\underline{F},\underline{m}}$, or simply P, defined recursively as follows:

- $E_0 = F_0 = \{0\}$.
- $E_0, \ldots, E_k$ being chosen, E_{k+1} is chosen as a random (in a unitary invariant way) subspace of F_{k+1} containing E_k. The condition of unitary

[1] In this study, a finite-dimensional subspace of a separable infinite-dimensional reference Hilbert space.

invariance can be rephrased as follows: E_{k+1} is the orthogonal direct sum of E_k with an $(\delta m)_k$-dimensional subspace of $E_k^{\perp_F}$ chosen with $\mu_{E_k^{\perp_F}}^{(\delta m)_k}$.

The recursive, Markovian, nature of this definition ensures that the appropriate analog of Kolmogorov's consistency criterion holds automatically, so that it extends to a probability measure on $\underline{m}$-Matryoshkas in $\underline{F}$, i.e., to infinite sequences.

More precisely, letting $\mathcal{E}_k$ denote the σ-algebra generated by the random subspaces $E_0, \ldots, E_k$ and $\sigma\big(\cup_{k\in\mathbb{N}}\mathcal{E}_k\big)$ the σ-algebra they generate, there is an unique $\sigma\big(\cup_{k\in\mathbb{N}}\mathcal{E}_k\big)$-probability measure on M which has the right finite-dimensional marginals. We complete this probability measure (with the σ-algebra $\sigma\big(\cup_{k\in\mathbb{N}}\mathcal{E}_k\big)$ suitably enlarged to a σ-algebra we denote by $\mathcal{E}$ to include null sets), leading to the complete filtered probability space $\big(M, \mathcal{E}, (\mathcal{E}_k)_{k\in\mathbb{N}}, P\big)$.

One basic question is under which conditions $\underline{E}$ is total (recall this means that $\cup_{k\in\mathbb{N}}E_k$ is dense in $\mathcal{H}$) with P-probability 1. Denoting by $\overline{E}$ the closure of $\cup_{k\in\mathbb{N}}E_k$ in $\mathcal{H}$, the question is whether $\overline{E}^{\perp}$ (which is also the orthogonal of $\cup_{k\in\mathbb{N}}E_k$) reduces to $\{0\}$ (with P-probability 1). This question has a simple and nice answer.

Fix an $\underline{m}$-Matryoshka $\underline{E}$ and let $v \in \mathcal{H}$. For $k \in \mathbb{N}$ denote by $U_k(v)$ (or simply U_k when there is no risk of confusion) the orthogonal projection of v on E_k (a finite-dimensional, hence closed, subspace of $\mathcal{H}$) and set $W_k = W_k(v) := v - U_k(v)$, the projection of v on the orthogonal complement of E_k in $\mathcal{H}$. Set $V_0(v) := 0$ and for $k \in \mathbb{N}^*$ set $V_k = V_k(v) := U_k(v) - U_{k-1}(v)$, which by construction is the orthogonal projection of v on the orthogonal complement of E_{k-1} in E_k. To summarize, for $k \in \mathbb{N}^*$, $v \in \mathcal{H}$ splits as a sum of three mutually orthogonal vectors

$$v = U_{k-1}(v) + V_k(v) + W_k(v)$$

corresponding to the orthogonal sum decomposition

$$\mathcal{H} = E_{k-1} \oplus \big(E_{k-1}^{\perp} \cap E_k\big) \oplus E_k^{\perp}.$$

Theorem 2.1. *For $k \in \mathbb{N}^*$ define $r_k := \frac{m_k - m_{k-1}}{n_k - m_{k-1}}$ if $n_k - m_{k-1} > 0$ and $r_k := 1$ otherwise.*

If the series $\sum_k r_k$ is divergent the space $\overline{E}^{\perp}$ is zero-dimensional (i.e., $\underline{E}$ is total) with probability 1.

If the series $\sum_k r_k$ is convergent the space $\overline{E}^{\perp}$ is infinite-dimensional with probability 1.

Lemma 2.2. *Let $v \in \mathcal{H}$.*

(i) As functions on $(M, \mathcal{E})$ with values in $\mathcal{H}$, $U_k(v)$, $V_k(v)$ and $W_k(v)$ are $\mathcal{E}_k$-measurable for each $k \in \mathbb{N}$.

(ii) The sequence $W_k(v)$ converges almost surely to 0 if and only if

$$\lim_{k\to\infty} \mathbb{E}\big(\|W_k(v)\|^2\big) = 0.$$

Corollary 2.3. *Let $i \in \mathbb{N}^*$ and $v \in F_i$. For $k \geq i$,*

$$\mathbb{E}\big(\|W_k(v)\|^2\big) = \mathbb{E}\big(\|W_{i-1}(v)\|^2\big) \prod_{j=i}^{k}(1 - r_j) \leq \|v\|^2 \prod_{j=i}^{k}(1 - r_j),$$

with equality if v belongs to the orthogonal complement of F_{i-1} in F_i.

Proof the first assertion of Theorem 2.1. By assumption, $\cup_{i\in\mathbb{N}}F_i$ is dense in $\mathcal{H}$. So $\cup_{k\in\mathbb{N}}E_k$ is dense in $\mathcal{H}$ if and only if it is dense in $\cup_{i\in\mathbb{N}}F_i$. Equivalently, $\cup_{k\in\mathbb{N}}E_k$ is dense in $\mathcal{H}$ if and only if any v in some F_i is in the closure of $\cup_{k\in\mathbb{N}}E_k$, i.e., if and only if the sequence $(W_k(v))_{k\geq i}$ converges to 0 for any v in some F_i.

Using the linearity of the map $v\mapsto W_k(v)$, to prove that $\lim_{k\to\infty}W_k(v)=0$ for every $v\in\cup_{i\in\mathbb{N}}F_i$, it suffices to check this convergence when v ranges over a basis of the finite-dimensional space F_i for each i. We fix an orthonormal basis $(v_n)_{n\geq 1}$ in $\mathcal{H}$ such that $F_i:=\mathrm{span}(v_n)_{n\in[\![1,n_i]\!]}$. Then $W_k(v_n))$ is well-defined for large enough k, i.e., for k such that $n_k\geq n$: $\cup_{k\in\mathbb{N}}E_k$ is dense in $\mathcal{H}$ if and only if $\lim_{k\to\infty}W_k(v_n)=0$ for every $n\in\mathbb{N}^*$.

By (ii) in Lemma 2.2 we know that $\lim_{k\to\infty}W_k(v)=0$ holds almost surely for a given v in some F_i if and only if $\lim_{k\to\infty}\mathbb{E}\left(\|W_k(v)\|^2\right)=0$. By the above remark, it is enough to let v range over the countable set $(v_n)_{n\geq 1}$.

By Corollary 2.3, for $i\in\mathbb{N}^*$, $n\in[\![n_{i-1}+1,n_i]\!]$ and $k\geq i$, $\mathbb{E}\left(\|W_k(v_n)\|^2\right)=\prod_i^k(1-r_j)$. Thus $\cup_{k\in\mathbb{N}}E_k$ is dense in $\mathcal{H}$ if and only if the infinite product "diverges" to 0, i.e., if $\lim_{k\to\infty}\prod_i^k(1-r_j)=0$ for every $i\in\mathbb{N}^*$. As $r_j\in[0,1]$ for each $j\in\mathbb{N}^*$, this is equivalent to the divergence of the series $\sum_j r_j$, concluding the proof. □

Proof the second assertion of Theorem 2.1. Our assumption now is that $\sum_k r_k$ is convergent. Hence, for large enough i, $s_i:=\prod_{k\geq i}(1-r_k)>0$, and $s_i\nearrow 1$ when $i\to\infty$.

By assumption, $\lim_{i\to\infty}n_i=\infty$. Thus $\mathbb{S}:=\{i\in\mathbb{N}^*,n_i-n_{i-1}>0\}$ is infinite. For each $i\in\mathbb{S}$ we choose a non-zero vector $v_i\in F_{i-1}^{\perp}\cap F_i$. By Corollary 2.3, $\mathbb{E}\left(\|W_k(v_i)\|^2\right)=\|v_i\|^2\prod_{j=i}^k(1-r_j)$ for $k\geq i$. Taking the large k limit, $\mathbb{E}\left(\|\Phi v_i\|^2\right)=\|v_i\|^2 s_i$ for $i\in\mathbb{S}$.

Fix $\varepsilon>0$, $k\in\mathbb{N}$ and $v\in F_k^{\perp}$. Write

$$\mathbb{E}\left(\|\Phi v\|^2\right)=\mathbb{E}\left(\|\Phi v\|^2\mathbf{1}_{A_k(\varepsilon)}\right)+\mathbb{E}\left(\|\Phi v\|^2\mathbf{1}_{A_k(\varepsilon)^c}\right).$$

On $A_k(\varepsilon)$, $\|\Phi v\|^2\leq\varepsilon^2\|v\|^2$ so

$$\mathbb{E}\left(\|\Phi v\|^2\mathbf{1}_{A_k(\varepsilon)}\right)\leq\varepsilon^2\|v\|^2P(A_k(\varepsilon)).$$

As $\|\Phi v\|^2\leq\|v\|^2$ (Φ is an orthogonal projector!)

$$\mathbb{E}\left(\|\Phi v\|^2\mathbf{1}_{A_k(\varepsilon)^c}\right)\leq\|v\|^2P\big(A_k(\varepsilon)^c\big).$$

Thus we have proven that, for $\varepsilon>0$, $k\in\mathbb{N}$ and $v\in F_k^{\perp}$,

$$\mathbb{E}\left(\|\Phi v\|^2\right)\leq\varepsilon^2\|v\|^2P(A_k(\varepsilon))+\|v\|^2(1-P(A_k(\varepsilon)).$$

For $i\in\mathbb{S}$, $i>k$, it holds that $v_i\in F_{i-1}^{\perp}\cap F_i\subset F_k^{\perp}$, so putting things together

$$\|v_i\|^2s_i=\mathbb{E}\left(\|\Phi v_i\|^2\right)\leq\big(\varepsilon^2-1\big)\|v_i\|^2P(A_k(\varepsilon))+\|v_i\|^2.$$

Dividing by $\|v_i\|^2$ (which is >0 by construction) and rearranging yields

$$\big(1-\varepsilon^2\big)P(A_k(\varepsilon))\leq 1-s_i,$$

and letting $i \to \infty$ along $\mathbb{S}$ we obtain that $P(A_k(\varepsilon)) = 0$ for every $\varepsilon < 1$, so $\cup_{k\in\mathbb{N}} A_k(\varepsilon) = \lim_{k\to\infty} A_k(\varepsilon)$ is a null set for every $\varepsilon < 1$.

Thus $A := \left\{\underline{E} \in M_{\underline{F},m} \text{ such that } G_{\underline{E}} \text{ is finite-dimensional}\right\}$, a subset of $\cup_{k\in\mathbb{N}} A_k(\varepsilon)$ is also a null set. To summarize, if $\sum_k r_k$ is convergent, the orthogonal $G_{\underline{E}}$ of $\cup_{k\in\mathbb{N}} E_k$ is infinite-dimensional with probability 1, concluding the proof. $\square$